Solution Architecture

System Components

Frontend

- Streamlit Web Application

Backend

- Python

Machine Learning Models

- BERT Emotion Classifier
- BiLSTM Emotion Classifier

AI Service

- Google Gemini API

Dataset

- Emotion Dataset

Storage

- CSV Session History

Workflow

User Input ↓

Text Preprocessing ↓

Emotion Detection (BERT & BiLSTM) ↓

Google Gemini AI ↓

Personalized Learning Suggestion ↓

Display Result

↓

Store Session History

Technologies Used

- Python
- Streamlit
- TensorFlow

- Transformers
- Google Gemini API
- Pandas
- NumPy